

Project Title: DSA Competitive Arena a Full-Stack Coding Practice Platform

Objective: The objective of this project is to build a functional Competitive Programming environment where users can browse algorithmic challenges, search for specific problems, and submit solutions to earn points. It aims to demonstrate how algorithmic logic (like searching and sorting) can be integrated directly into a web application's backend.

Features: The DSA Competitive Arena is a specialized platform for algorithmic practice featuring a dynamic problem set with challenges ranging from binary search to dynamic programming. It includes an optimized search engine using binary search logic to find problems instantly and a real-time leaderboard that utilizes merge sort to rank participants by their scores. Users can submit solutions through a dual-submission system (text input or file upload) and access an automated documentation viewer for reference materials. The entire experience is delivered through a modern, darkmode UI designed specifically for developer productivity.

Applications in Real Life: The DSA Competitive Arena functions as a professional Technical Assessment Portal for corporate hiring and a Learning Management System for educational institutions to host coding challenges. It demonstrates the Scalable Logic used by platforms like LeetCode to handle large datasets through optimized searching and sorting. By automating submissions and rankings, it serves as a high-efficiency Performance Analytics tool for tutoring centers and coding bootcamps.

Algorithms Used: Binary search, Merge sort.

Tools Used: Python, Flask, OS module, Jinja2, HTML, CSS